Nuclear Astrophysics

Exercise 7

In this exercise, we will solve the Tolman-Oppenheimer-Volkov (TOV) equations, following the instructions of Silbar & Reddy (2004) `https://arxiv.org/pdf/nucl-th/0309041.pdf`. You can find it in moodle under 'Neutron stars for undergrad'. Use the supplementary python script 'mytov.py'.

a) Read the paper and determine the equations of state (EoS) for:

   i) a white dwarf (WD), modelled as a relativistic degenerate electron gas,

   ii) a pure neutron star (NS), modelled as a noninteracting Fermi gas (neutrons) for "arbitrary" relativity,

   iii) a realistic NS with simplified EoS that takes nuclear interactions into account.

   Open the python script and insert the corresponding coefficients and exponents for the different EoS in the function `eos(p,case)`. Run the program to get plots of the three mass-radius relations.

b) What do you find for (relativistic) WD of arbitrary volume?

c) Compare the results for the two NSs:

   - What is the maximum mass in both cases? What happens left of the maximum (i.e. for smaller radii)?
   - Why are the maximum masses different?
   - Which NS masses do we observe in nature?